

Chemistry Lab 4

WATER SPLITTING EXPERIMENT

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Abstract - In this experiment, we will analyse the process of splitting water using electrolysis under four conditions: DI water, Tap water, Added NaCl and Added MgCl₂. The amount of hydrogen released was also analysed, and the relative rate of the reaction under these conditions was studied.

I. INTRODUCTION

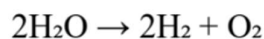
In this experiment, we will explore the process of electrolysis. In electrolysis, we use electric current to perform a chemical reaction. When we use electrolysis on water, oxidation occurs at the anode and oxygen is collected there. Similarly, the reduction of water occurs at the cathode and hydrogen is produced there. This is a very clean and cheap method of producing Hydrogen gas, which has multiple applications, such as in fuel. The rate and efficiency of this reaction, is however, heavily dependent on the type of electrolyte and electrode used.

II. MATERIALS

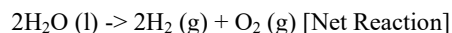
Copper Wire, Graphite Rods, graduated burette, Battery (9V PP3), Beaker, Electrolytes.

III. THEORY AND CONCEPTS

As discussed previously, in the electrolysis of water, oxidation occurs at the anode and oxygen is collected there, while the reduction of water occurs at the cathode and hydrogen is produced there. The balanced reaction is as given below:



However, we have different reactions happening at the anode and cathode. At the anode,
 $2\text{H}_2\text{O} (\text{l}) \rightarrow \text{O}_2 (\text{g}) + 4\text{H}^+ (\text{aq}) + 4\text{e}^-$ [Reaction at Anode]
 $2\text{H}_2\text{O} (\text{l}) + 2\text{e}^- \rightarrow \text{H}_2 (\text{g}) + 2\text{OH}^- (\text{aq})$ [Reaction at Cathode]



It has also been observed that the rate of reaction, which is in turn determined by the quantity of hydrogen gas evolved, significantly increases because of the increase in the concentration of charged ions in the water. We will also confirm these statements in the results and observations section.

IV. PROCEDURE

- For Tap water, we first filled the beaker with Tap water till ~150ml.

- The electrodes were placed in that beaker inside a water-filled inverted burette.
- The water was filled inside the burettes using a small hand-held pump, which created a vacuum inside the burette.
- The electrodes were then connected to the battery, and the current was passed.
- The setup was kept under observation for about 5 minutes.
- The Tap water was then discarded DI water was added, and the above procedure was repeated again.
- Then, the solution of electrolyte was created using the DI water (150ml) and the given solution of NaCl was added. The above procedure was again done for 30 minutes after the level of hydrogen reached beyond the zero mark and the level observations were taken for every 3 minutes.
- The NaCl solution was discarded, and a new solution of MgCl₂ in DI was added. The same process was done for NaCl.

V. RESULTS AND OBSERVATIONS

Tap Water: There was some amount of bubbling visible at the electrodes, resulting in the evolution of H₂ gas. However, the amount of H₂ gas collected at the end of 5 minutes was too less for practically measuring.

DI Water: There was an absolutely negligible amount of bubbling visible at the electrodes. There as also no hydrogen gas collected at the top of the Measuring cylinder.

NaCl and MgCl₂ Solutions: The amount of Hydrogen gas released was quite significant and in a measurable quantity. The tabulated quantities of the Hydrogen gas released in both cases are as tabulated below. The corresponding plot is also added below:

Time (min)	Level of H ₂ (ml) with:	
	NaCl	MgCl ₂
0	0	0
3	0.8	1
6	1.6	2
9	2.2	2.8
12	3.2	3.8
15	4.2	4.8
18	5.2	5.6
21	6.2	6.4

24	7.2	7.4
27	8	8.2
30	8.8	9.2

Table 1. Amount of H_2 gas evolved with NaCl and $MgCl_2$ against time.

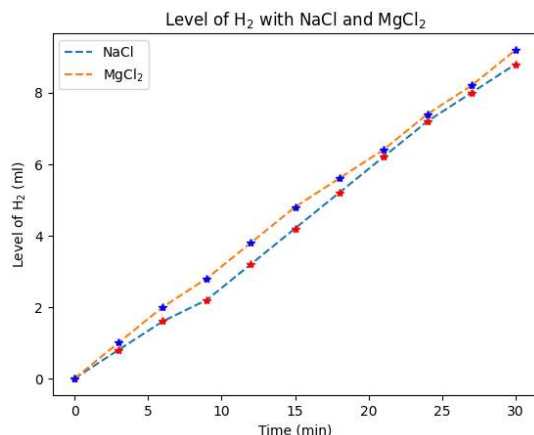


Fig. 1. Plot for amount of H_2 gas evolved with NaCl and $MgCl_2$ against time.

A. Main Observations

Starting with the tap water, a major observation is that there was some amount of bubbling visible, but it was not so rapid that a significant amount of Hydrogen gas could be collected. This can be attributed to some small quantity of ionisation that could be happening in the tap water. On the other hand, in the DI water, there was absolutely no bubbling visible. This can be attributed to the fact that the DI water is completely de-ionised and therefore, there is no bubbling happening. In the case of both NaCl and $MgCl_2$, rapid bubbling was seen with a substantial amount of hydrogen gas being collected. This quantity is given in the table above. From the plot, we can also observe that the amount of hydrogen evolved in NaCl is slightly less than that for $MgCl_2$. This can be possibly due to the fact that upon ionisation, NaCl gives rise to two ions, namely Na^+ and Cl^- . However, the ionisation of $MgCl_2$ gives rise to three ions, Mg^{+2} and $2Cl^-$. This, in total, gives more ions and a more net charge, therefore carrying greater charges and finally leading to the passage of a greater amount of current, finally leading to a greater production of Hydrogen. We also observed that the amount of Hydrogen released was much greater than the amount of Oxygen released. There are two possible reasons for this; the first one is that, upon the electrolysis of two moles of water, we obtain two moles of hydrogen but only one mole of

oxygen. The second reason might be attributed to the fact that the solubility of oxygen in water is much greater than that of hydrogen, as we found from the source [2]. So, more amount of oxygen was being dissolved into the water compared to hydrogen and the corresponding results were seen.

VI. REFERENCES

- [1] Lab Manual
[2] Kolev, N.I. (2011). Solubility of O_2 , N_2 , H_2 and CO_2 in water. In: Multiphase Flow Dynamics 4. Springer, Berlin, Heidelberg.
https://doi.org/10.1007/978-3-642-20749-5_11

VII. AUTHOR CONTRIBUTIONS

Jaidev	Experimentation, Report Development
Pranav	Experimentation, Report Review
Sibtain	Experimentation, Report Review
Suteekshna	Experimentation, Report Review

VIII. RAW DATASET

Time	NaCl	MgCl ₂
0	0	0
3	0.8	1
6	1.6	2
9	2.2	2.8
12	3.2	3.8
15	4.2	4.8
18	5.2	5.6
21	6.2	6.4
24	7.2	7.4
27	8.0	8.2
30	8.8	9.2

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